

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	Ruthika Reddy	Reg. No.:	192421138
Section / Batch:	Btech IT	Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze the given questions thoroughly before answering.
- Compare multiple aspects such as methods, results, advantages, and limitations clearly.
- Critically evaluate the strengths, weaknesses, and implications with proper justification.
- Present meaningful insights, interpretations, and well-supported conclusions from the comparisons.

Question Type	Focus Area	Questions
Comparative Analysis	Comparative analysis of Dynamic Programming and Greedy algorithms for optimization problems.	Q1

SECTION A – Comparative Analysis

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Comparative Analysis

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%				
Comparison & Differentiation	25%				
Critical Evaluation	25%				
Synthesis & Insight Generation	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
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Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



SIMATS Online Programming Lab

DAA PROGRAMMING



C RUTHIKA REDDY
192421138RA

Shortest Path vs MST

100 marks

1/1 answered

Given a graph, compute shortest paths using Dijkstra and MST using Prim's algorithm. Compare selection criteria and outputs. Analyze differences in objectives. Evaluate complexity. Discuss applicability. Generate insights on graph algorithms.

Your Code

```
1 import heapq
2
3 n = int(input())
4
5 graph = []
6 for i in range(n):
7     graph.append(list(map(int, input().split())))
8
9 src = int(input())
10
11 # Dijkstra
12 dist = [999] * n
13 dist[src] = 0
14 pq = [(0, src)]
15
16 while pq:
17     d, u = heapq.heappop(pq)
18
```

```

19     if d != dist[u]:
20         continue
21
22     for v in range(n):
23         if graph[u][v] != 0:
24             nd = d + graph[u][v]
25
26             if nd < dist[v]:
27                 dist[v] = nd
28                 heapq.heappush(pq, (nd, v))
29
30 print("Dijkstra Shortest Paths:")
31 for i in range(n):
32     print(src, "to", i, "=", dist[i])
33
34
35 # Prim
36 visited = [False] * n
37 visited[0] = True
38 total = 0
39
40 print("Prim MST:")
41
42 for i in range(n - 1):
43     minimum = 999
44     x = -1
45     y = -1
46
47     for u in range(n):
48         if visited[u]:
49             for v in range(n):
50                 if not visited[v] and graph[u][v] < minimum:
51                     minimum = graph[u][v]
52                     x = u
53                     y = v
54
55     if x != -1:
56         print(x, "-", y, "=", minimum)
57         total += minimum
58         visited[y] = True
59
60
61 print("Total MST cost =", total)
62

```

Input

```
2 0 3 5
6 3 0 1
0 5 1 0
0
```

Output

```
PRIM MST:
0 - 1 = 2
1 - 2 = 3
2 - 3 = 1
Total MST cost = 6
```

 Run Save Submit

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